



Exploratory Data Analysis

Discuss basics of multivariate statistical analysis

Carry out hierarchical clustering of samples

Discuss and perform principal component analyses

Produce 'small multiples' plot

Use standard dplyr 'verbs' to query our data

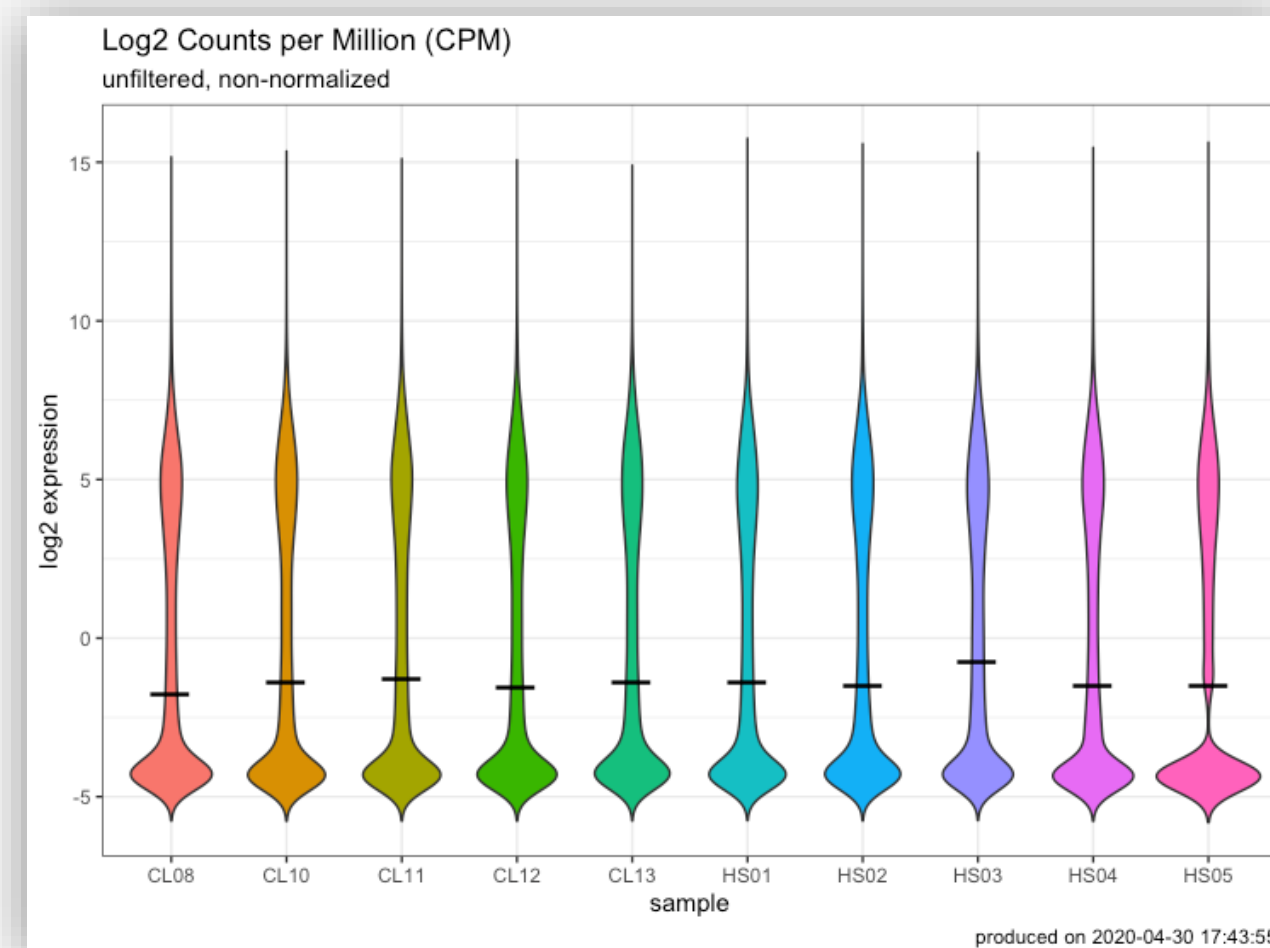
Produce interactive graphics using plotly

Produce interactive tables with DT package

What we accomplished in the last lecture

Raw

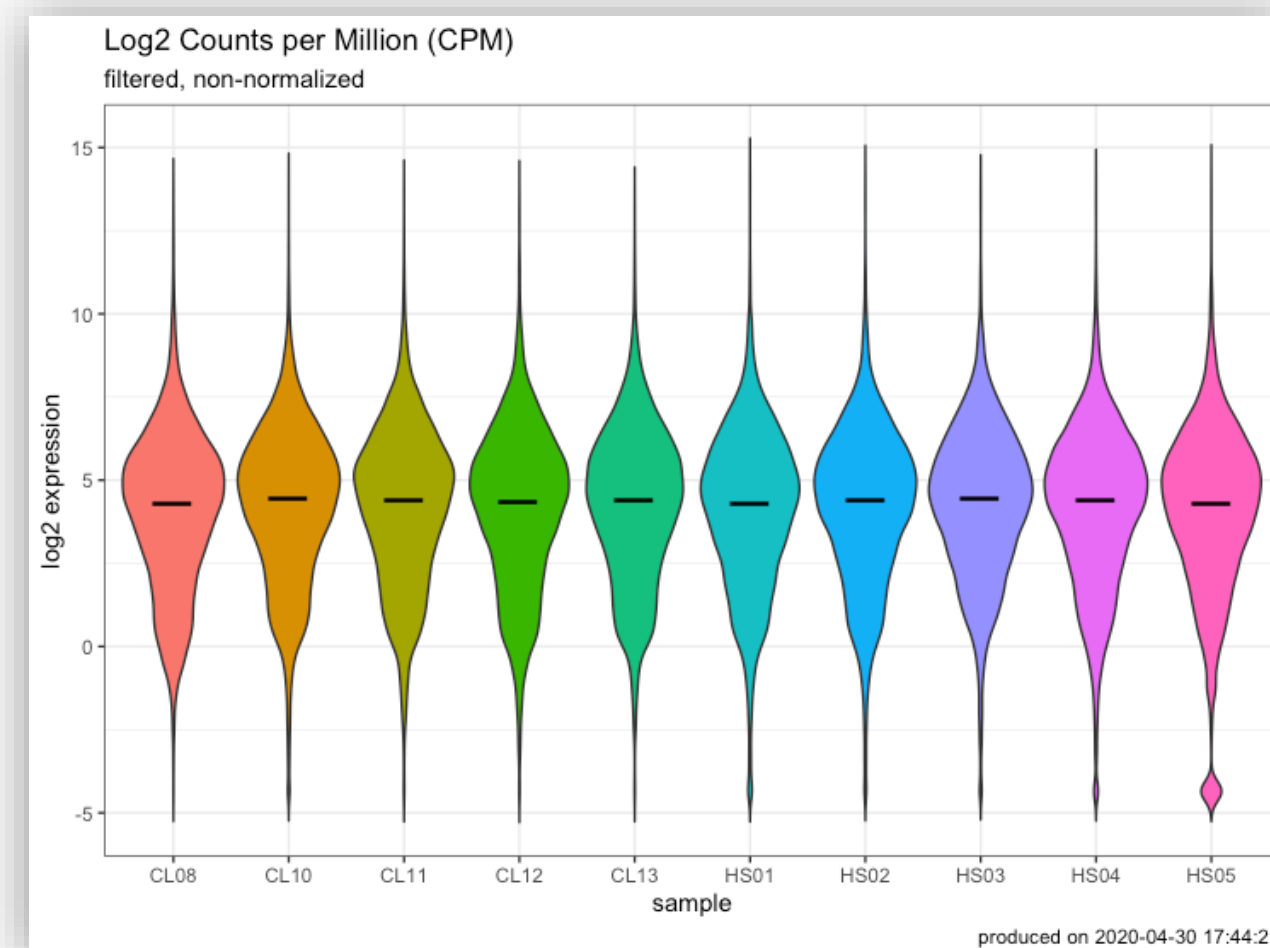
as_tibble()
pivot_longer()
ggplot()



Filtered

Base R
rowSums()

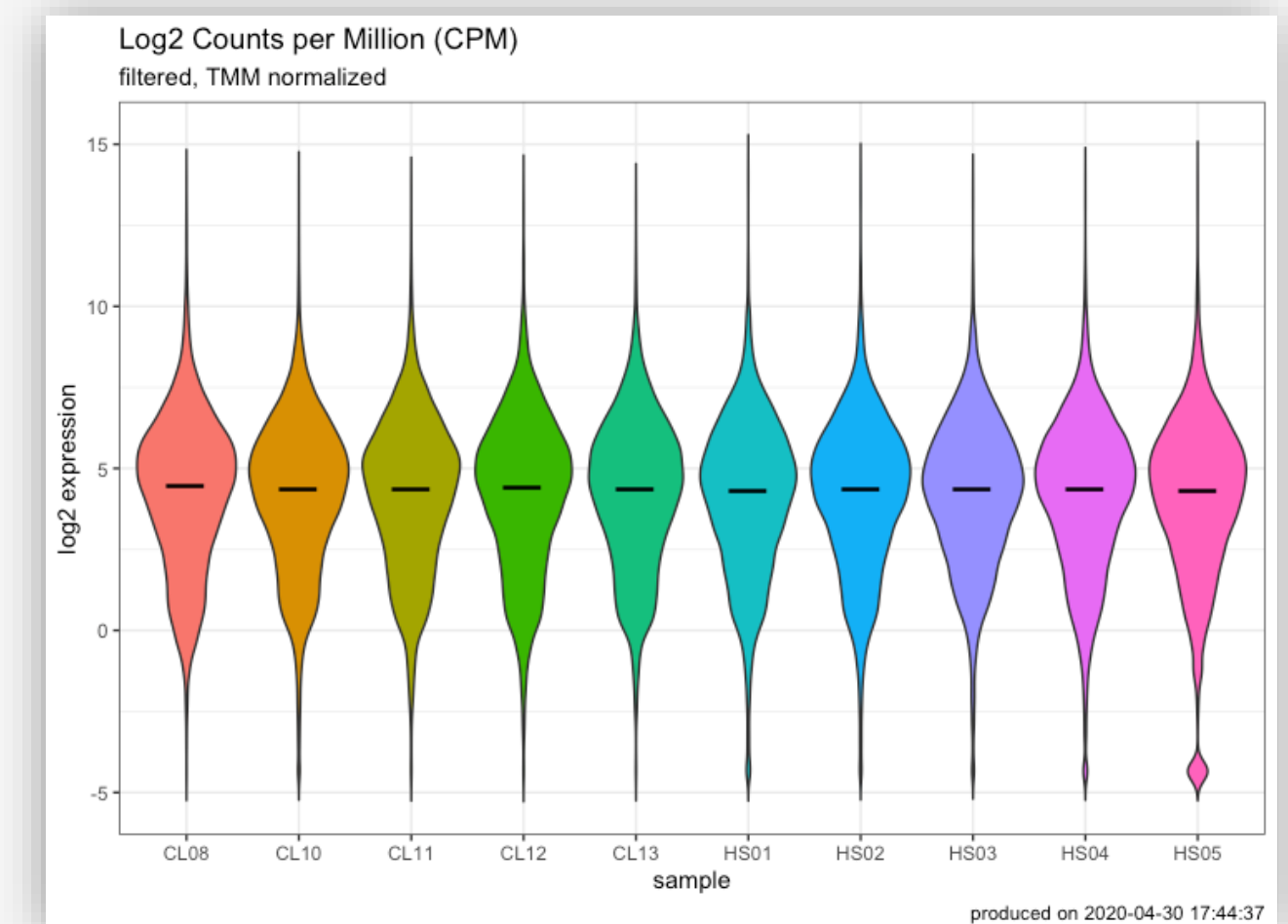
as_tibble()
pivot_longer()
ggplot()



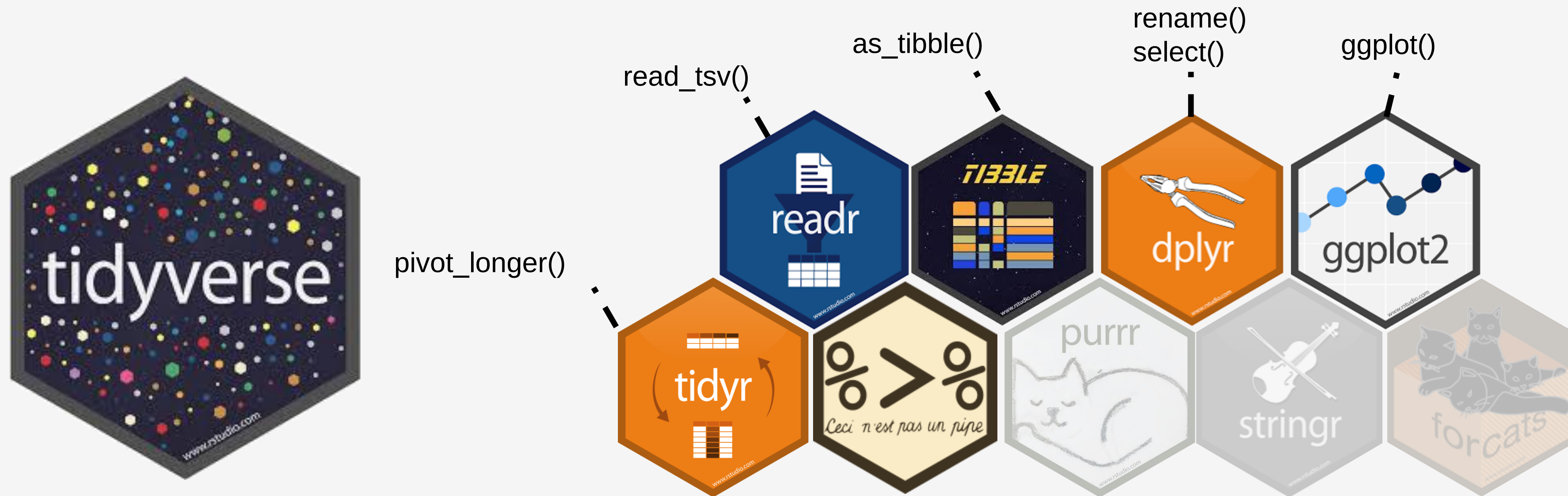
Normalized

edgeR::
calcNormFactors

as_tibble()
pivot_longer()
ggplot()



The Tidyverse

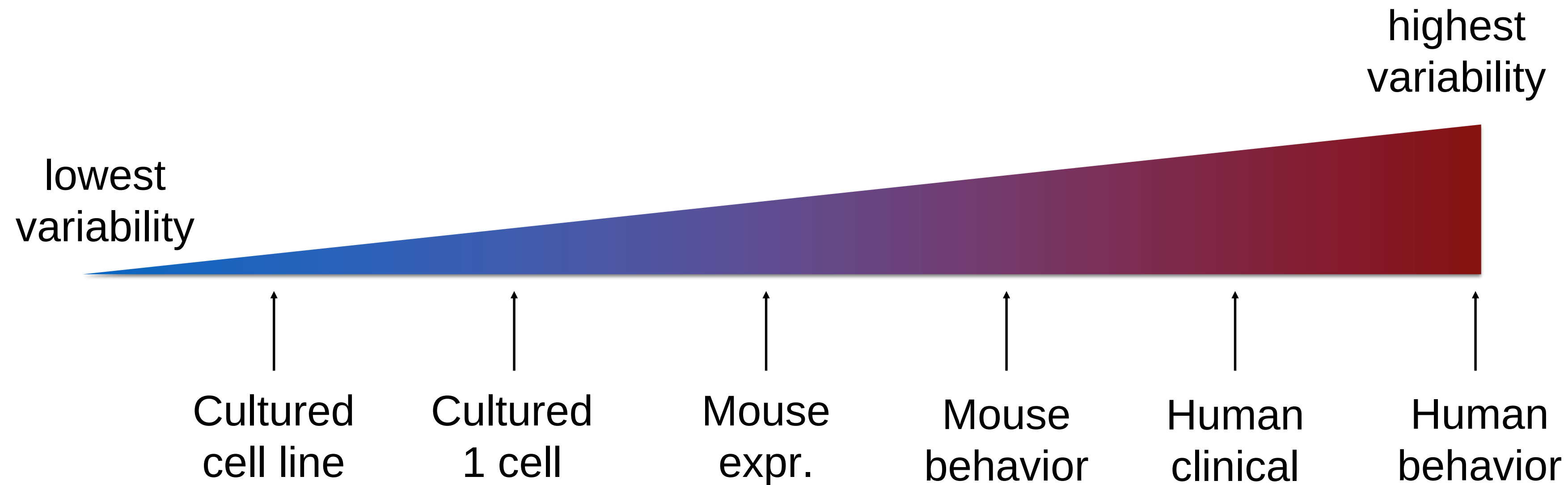


- Tidyverse packages/functions only work with data frames
- The readr package always produces a data frame
- A tibble is still a data frame, but with nicer display and subsetting
- ‘Piping’ with %>% (magrittr package) makes code more succinct and readable
- Once you grasp the layered approach of ggplot, easy to connect with other R packages (e.g., Plotly and gt)

Understanding sources of variation

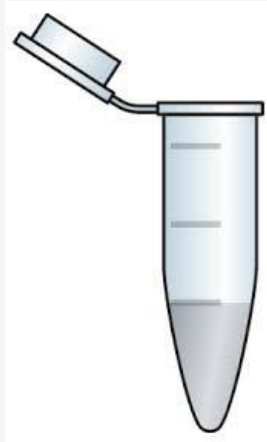
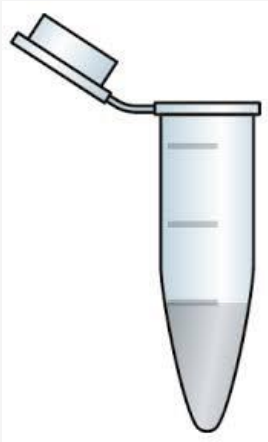
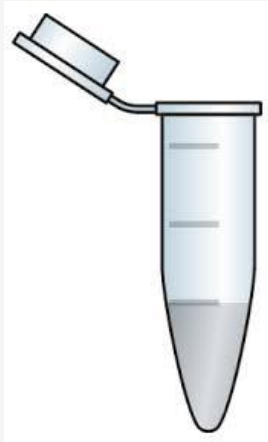
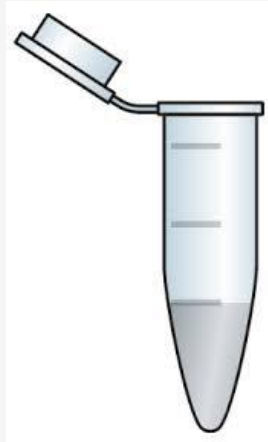
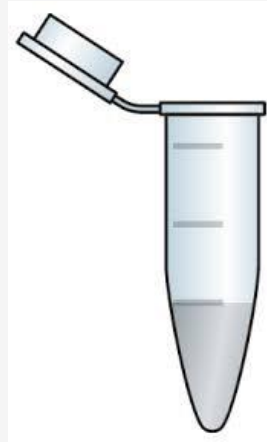
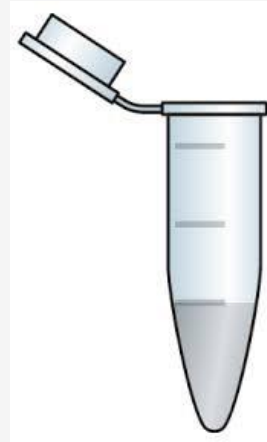
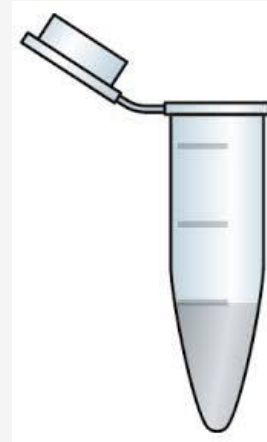


When it comes to potential sources of variance, not all experiments are created equal



The type of experiment will influence your sources of variance and the types of metadata you need to consider

Potential sources of variation

variable of interest								
	wt	wt	wt	wt	ko	ko	ko	ko
								
sex	♂	♂	♀	♂	♀	♀	♂	♂
batch	1	1	1	1	1	1	2	2
batch	1	2	3	4	1	2	3	4
run	1	1	1	1	1	1	1	1
person	1	2	1	1	1	2	1	1
x	1	1	1	1	2	2	2	2
confounding variables								

‘Metadata’ = data about your data

**How do we visualize variance in
our data?**

**If a variable is not recorded
....you can't!**

**How do we visualize variance in
our data?**

Many options for dimensional reduction

Method	Input Data	Method Class	Nonlinear
PCA	continuous data	unsupervised	
CA	categorical data	unsupervised	
MCA	categorical data	unsupervised	
PCoA (cMDS)	distance matrix	unsupervised	
NMDS	distance matrix	unsupervised	
Isomap	continuous*	unsupervised	✓
Diffusion Map	continuous*	unsupervised	✓
Kernel PCA	continuous*	unsupervised	✓
t-SNE	continuous/distance	unsupervised	✓
Barnes–Hut t-SNE	continuous/distance	unsupervised	✓
LDA	continuous (X and Y)	supervised	
PLS (NIPALS)	continuous (X and Y)	supervised	
NCA	distance matrix	supervised	✓
Bottleneck NN	continuous/categorical	supervised	✓
STATIS	continuous	multidomain	
DiSTATIS	distance matrix	multidomain	

Many options for dimensional reduction

Method	R function
PCA	<code>stats::prcomp</code>
CATPCA	<code>gifi::princals</code>
CA	<code>FactoMineR::CA</code>
MCA	<code>FactoMineR::MCA</code>
PCoA (cMDS)	<code>stats::cmdscale</code>
NMDS	<code>ecodist::nmDS</code>
Isomap	<code>vegan::isomap</code>
Diffusion Map	<code>diffusionMap::diffuse</code>
(Barnes–Hut) t-SNE	<code>Rtsne::Rtsne</code>
LDA	<code>MASS::lda</code>
PLS (NIPALS)	<code>mixOmics::pls</code>
DiSTATIS	<code>DistatisR::distatis</code>
Procrustes	<code>vegan::procrustes</code>

Understanding PCA

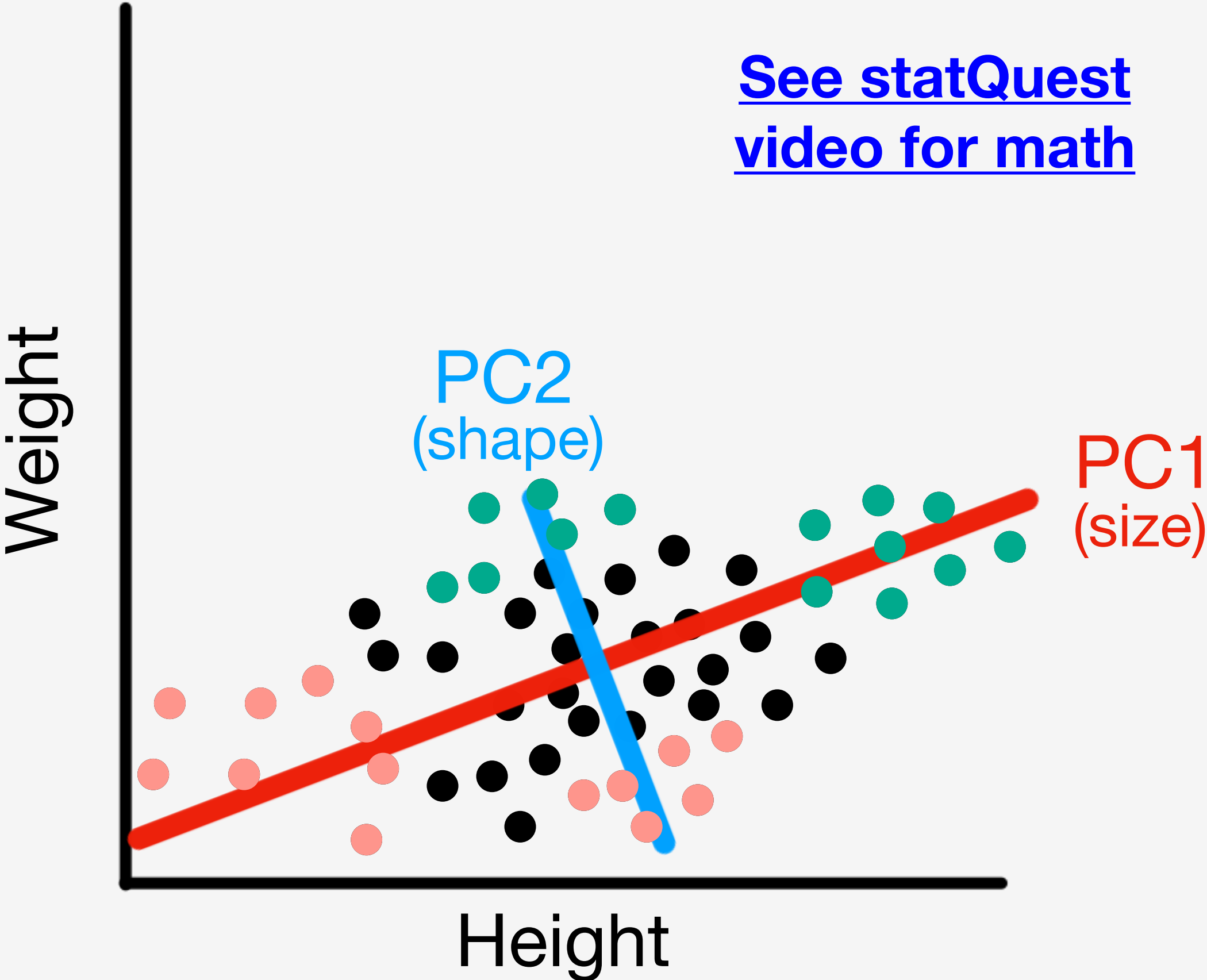
2D

	Person 1	...	Person n
Height	72	60	75
Weight	150	190	130

Multi-D

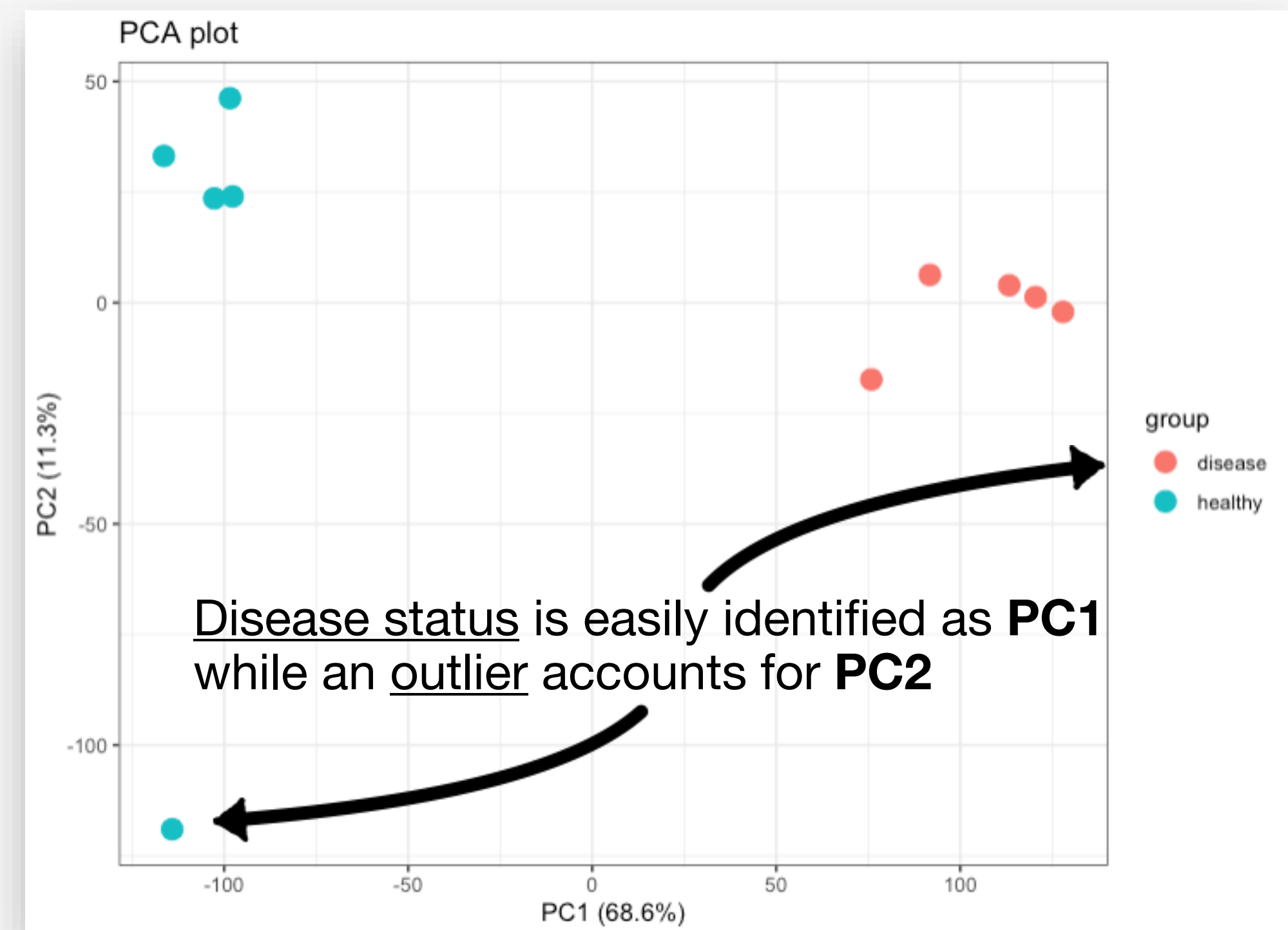
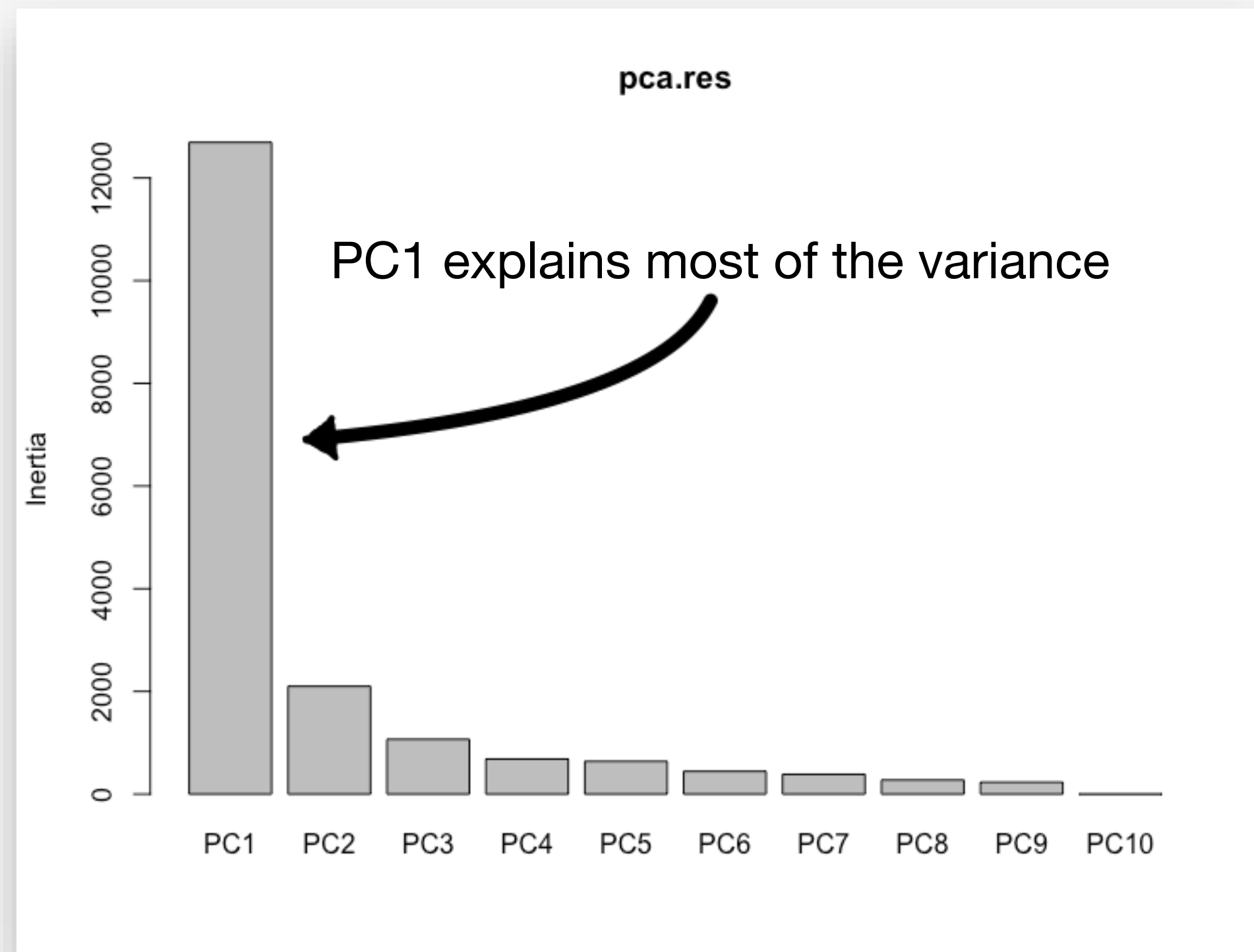
	Person 1	...	Person n
Gene A	72	60	75
Gene B	150	190	130
Gene C	72	60	75
Gene D	150	190	130
Gene E	72	60	75
Gene F	150	190	130
Gene G	72	60	75
...	150	190	130
Gene n	72	60	75

[See statQuest
video for math](#)



Run our PCA and plot the results
step 3 script

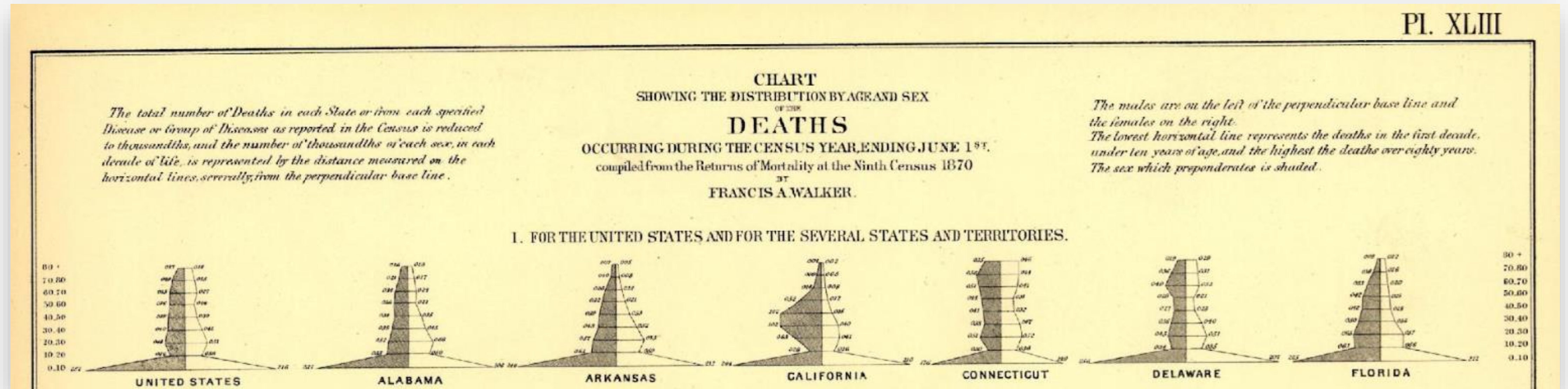
What if it's not so simple?



You have many covariates to explore and substantial variation explained by multiple PCs

‘Small multiples’ plot

Similar graphs using the same axes and labels



PC1

PC2

PC3

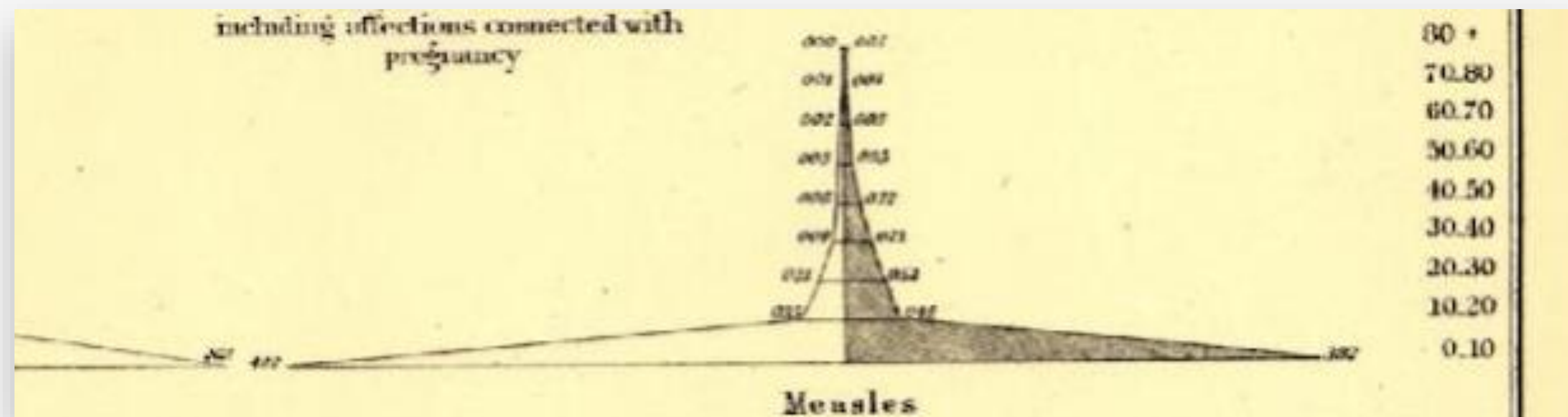
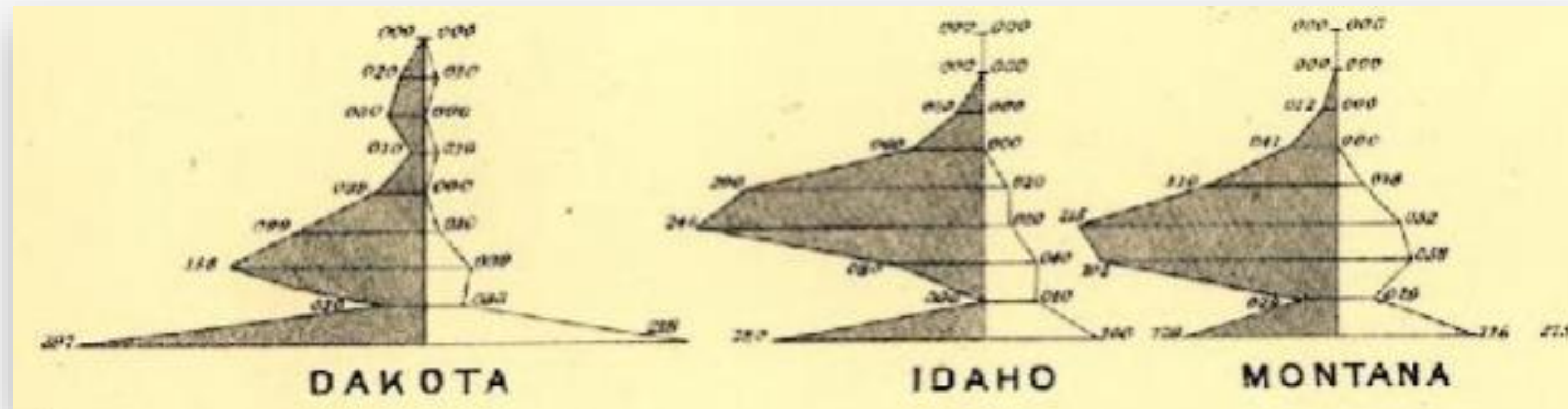
PC4

PC5

PC6

PC7

small multiples make it easy to spot interesting combinations of variables



back to script

**Take the time to thoroughly explore your
data before downstream analyses**

Using %>% to streamline your code

```
my_new_object <- doSomething(my_data)
my_new2_object <- doSomething_else(my_new_object)
my_new3_object <- doSomething_different(my_new2_object)
```



Magrittr package

```
my_new_object <- my_data %>%
  doSomething() %>%
  doSomething_else() %>%
  doSomething_different()
```

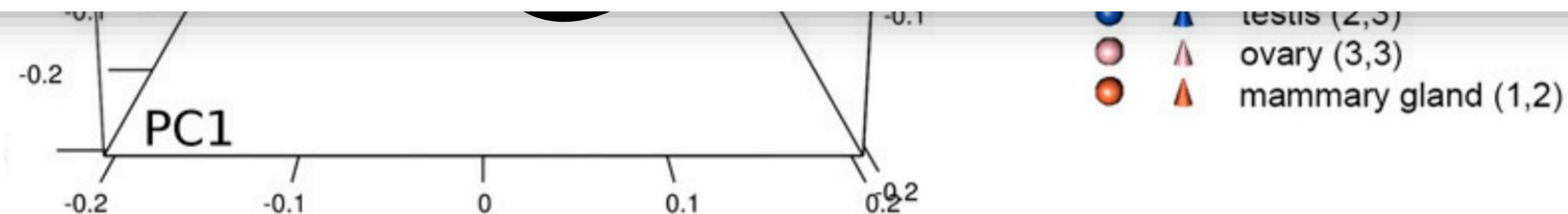

Comparison of the transcriptional landscapes between human and mouse tissues

Shin Lin^{a,b,1}, Yiing Lin^{c,1}, Joseph R. Nery^d, Mark A. Urich^d, Alessandra Breschi^{e,f}, Carrie A. Davis^g, Alexander Dobin^g, Christopher Zaleski^g, Michael A. Beer^h, William C. Chapman^c, Thomas R. Gingeras^{g,i}, Joseph R. Ecker^{d,j,2}, and Michael P. Snyder^{a,2}

^aDepartment of Genetics, Stanford University, Stanford, CA 94305; ^bDivision of Cardiovascular Medicine, Stanford University, Stanford, CA 94305; ^cDepartment of Surgery, Washington University School of Medicine, St. Louis, MO 63110; ^dGenomic Analysis Laboratory, The Salk Institute for Biological Studies, La Jolla, CA 92037; ^eCentre for Genomic Regulation and UPF, Catalonia, 08003 Barcelona, Spain; ^fDepartament de Ciències Experimentals i de la Salut, Universitat Pompeu Fabra, 08003 Barcelona, Spain; ^gFunctional Genomics, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11742; ^hMcKusick-Nathans Institute of Genetic Medicine and the Department of Biomedical Engineering, Johns Hopkins University, Baltimore, MD 21205; ⁱAffymetrix, Inc., Santa Clara, CA 95051; and ^jHoward Hughes Medical Institute, The Salk Institute for Biological Studies, La Jolla, CA 92037

Significance

To date, various studies have found similarities between humans and mice on a molecular level, and indeed, the murine model serves as an important experimental system for biomedical science. In this study of a broad number of tissues between humans and mice, high-throughput sequencing assays on the transcriptome and epigenome reveal that, in general, differences dominate similarities between the two species. These findings provide the basis for understanding the differences in phenotypes and responses to conditions in humans and mice.





RESEARCH ARTICLE

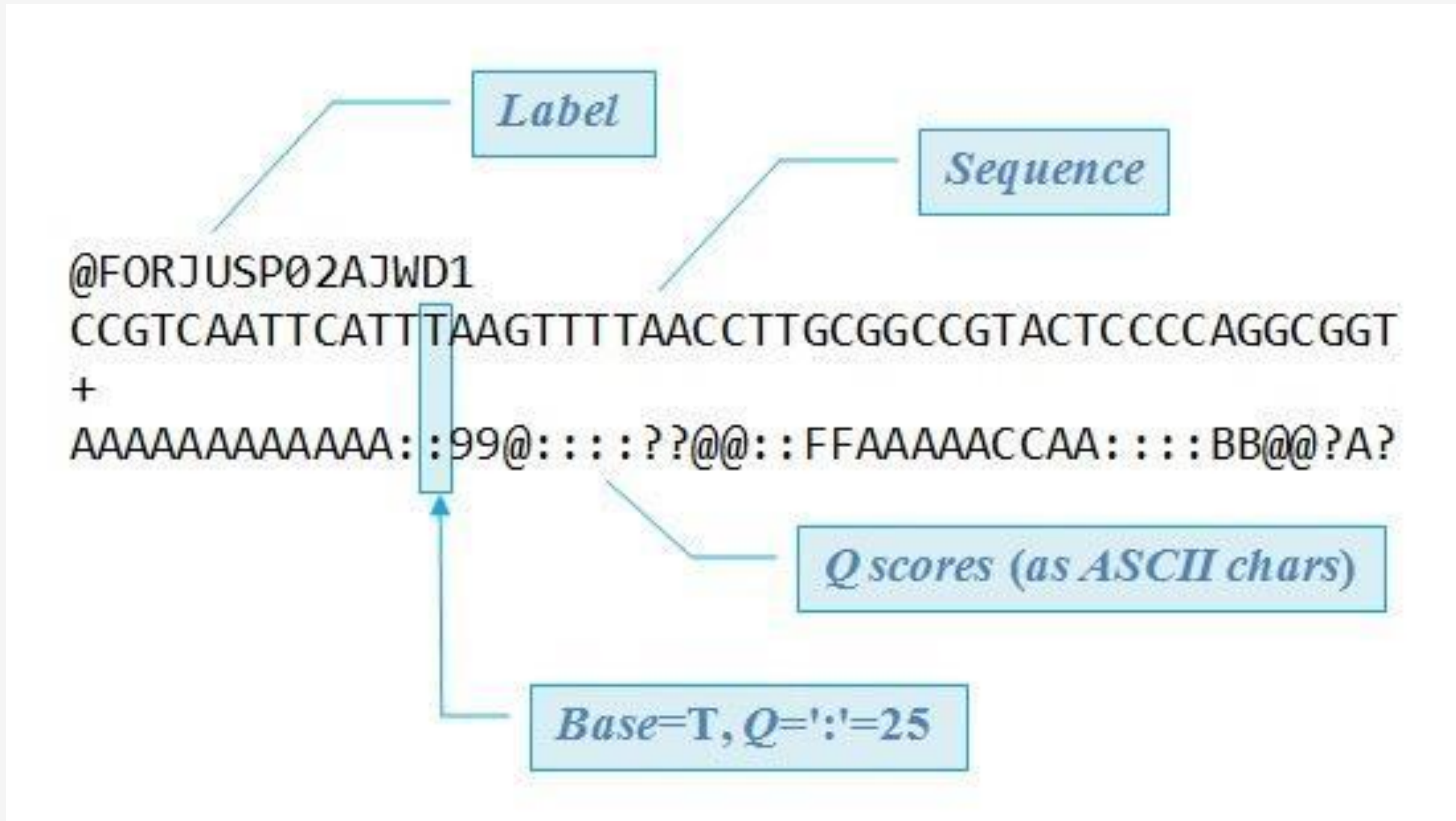
A reanalysis of mouse ENCODE comparative gene expression data [v1; ref status: awaiting peer review, <http://f1000r.es/5ez>]

Yoav Gilad, Orna Mizrahi-Man

Department of Human Genetics, University of Chicago, Chicago, IL, 60637, USA

“The implications of the observation that human and mouse gene expression data may be clustering by species more than by tissues can be profound. To a large degree, modern biology is built upon the empirical observation that homologous gene regulatory networks establish the identities of homologous cell-types, tissues, and organs across species – the results of Lin *et al.*, if true, challenge these observations and the biological basis of homology. From a more practical perspective, the mouse is arguably the most important animal model for biomedical research. If gene regulation in any mouse tissue is markedly more representative of a general mouse regulatory network than the regulatory network of a corresponding human tissue, this would call into question the utility of the mouse, and perhaps any other non-human animal, as a useful model system for biomedical research.”

remember the structure of a .fastq file?



Machine ID **@D4LHBFN1** run# **:276** Flowcell ID **:C2HKJACXX** Lane **:4** Tile **:1101** Xpos **:3448** Ypos **:12374** Read **1** Is filtered? **N** control# **:0** Barcode **:AGTTCC**

D87PMJN1 (run 253, flow cell D2GUAACXX, lane 7)	D87PMJN1 (run 253, flow cell D2GUAACXX , lane 8)	D4LHBFN1 (run 276, flow cell C2HKJACXX , lane 4)	MONK (run 312, flow cell C2GR3ACXX , lane 6)	HWI-ST373 (run 375, flow cell C3172ACXX , lane 7)	
heart	adipose	adipose	heart	brain	
kidney	adrenal	adrenal	kidney	pancreas	
liver	sigmoid colon	sigmoid colon	liver	brain	
small bowel	lung	lung	small bowel	spleen	
spleen	ovary	ovary	testis		Human
testis		pancreas			Mouse

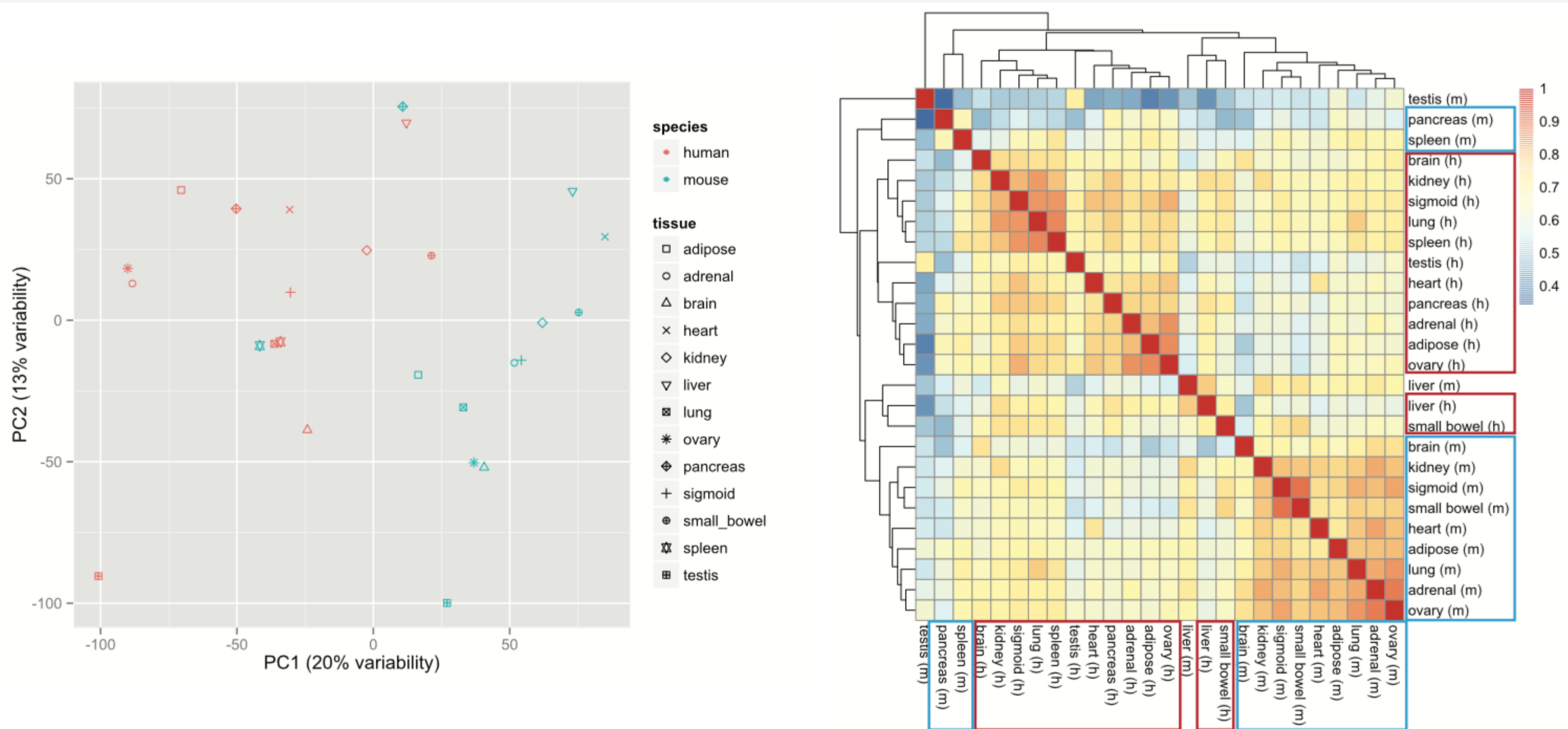
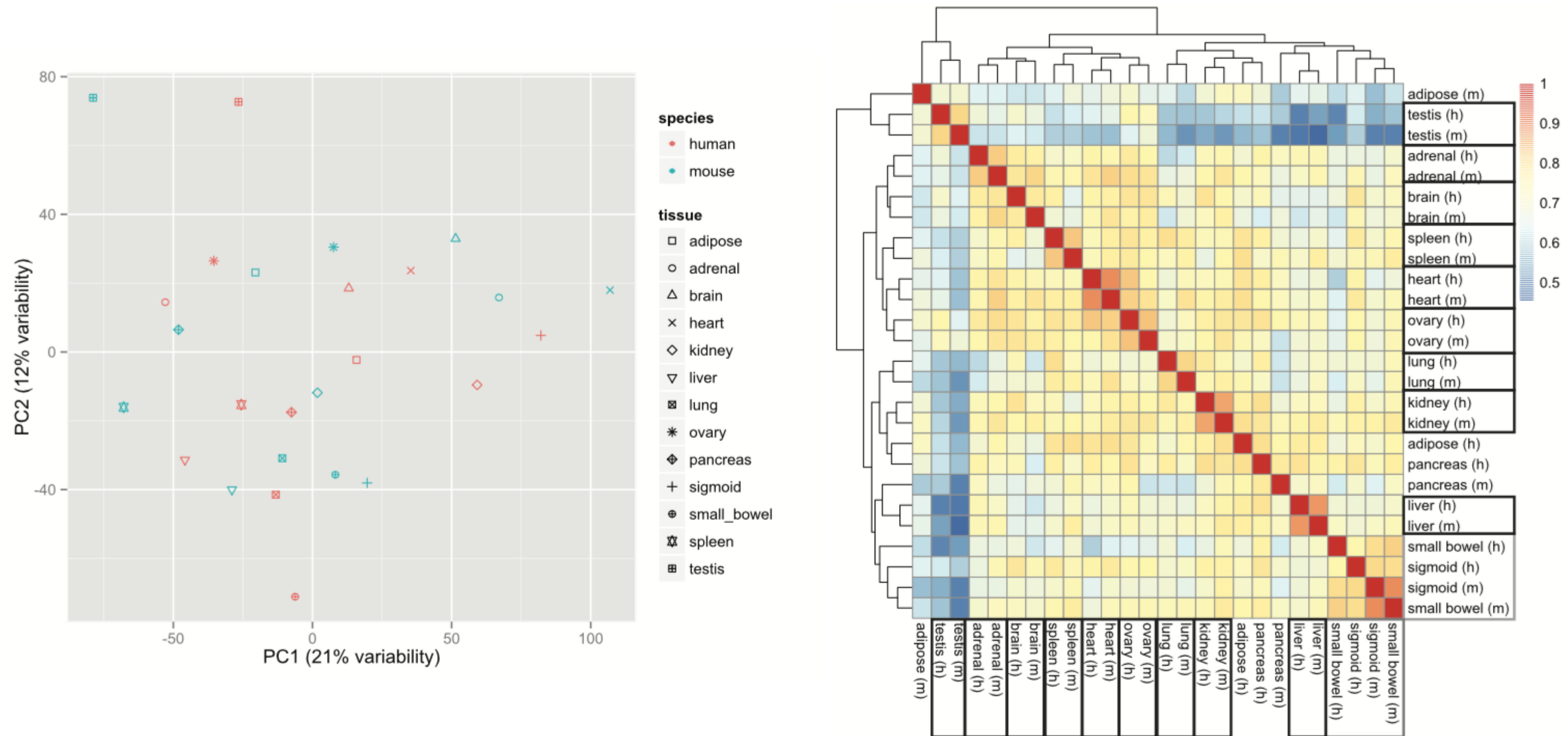


Figure 2. Recapitulating the patterns reported by the mouse ENCODE papers. **a.** Two-dimensional plots of principal components calculated by performing PCA of the transposed log-transformed FPKM values (from 14,744 orthologous gene pairs) for the 26 samples, after removal of invariant columns (genes). **b.** Heatmap based on pairwise Pearson correlation of expression data used in panel **a**. We used Euclidean distance and complete linkage as distance measure and clustering method, respectively.



Discuss this Article

Reader Comment 20 May 2015

Shin Lin, Department of Genetics, Stanford University, USA

“There remains the issue of our study design with respect to confounding of lane effect and species. It should be noted that our study design minimized library preparation and primer index effect. Although our experience is consistent with [reports] showing that lane effect is not a large contributor to variance, we recognize it is better to have data. Thus, we are sequencing under a new pattern of pooled libraries, and soon, we will post the results for the community.” - *authors of PNAS study*

Author Response 20 May 2015

Yoav Gilad, Human Genetics, University of Chicago, USA

Why are you calling these 'lane effects'? The samples were sequenced on different instruments (different sequencers) and by default - different flow cells. Can you explicitly acknowledge that fact please?

**This Issue**Views **24,937** | Citations **0** | Altmetric **553****Comment & Response****FREE**

October 8, 2019

More ▾

Notice of Retraction. Aboumatar et al. Effect of a Program Combining Transitional Care and Long-term Self-management Support on Outcomes of Hospitalized Patients With Chronic Obstructive Pulmonary Disease: A Randomized Clinical Trial. *JAMA*. 2018;320(22):2335-2343.

Hanan Aboumatar, MD, MPH^{1,2,3,4,5}; Robert A. Wise, MD⁶[» Author Affiliations](#) | [Article Information](#)*JAMA*. 2019;322(14):1417-1418. doi:10.1001/jama.2019.11954

The identified programming error was in a file used for preparation of the analytic data sets for statistical analysis and occurred while the variable referring to the study "arm" (ie, group) assignment was recoded. The purpose of the recoding was to change the randomization assignment variable format of "1, 2" to a binary format of "0, 1." However, the assignment was made incorrectly and resulted in a reversed coding of the study groups. Even though the data analyst created and conducted some test analysis programs, they were of the type that did not show any labeling of the arm categories, only the "arm" variable in a regression, for example. After detecting this

back to script

Data wrangling with the dplyr package

Four key 'verbs' of dplyr

1. **Mutate** - add new columns as a function of existing data
2. **Filter** - picks rows based on rules you apply
3. **Select** - picks columns based on rules you apply
4. **Arrange** - reorders rows by sorting based on column(s)

```

181 myData <- mutate(resultTable.unfiltered,
182   #average reps for the male LEPZQ at each timepoint
183   Mctl_LEPZQ_AVG = (Mctl_LEPZQ_1 + Mctl_LEPZQ_2 + Mctl_LEPZQ_3)/3,
184   M1h_LEPZQ_AVG = (M1h_LEPZQ_1 + M1h_LEPZQ_2 + M1h_LEPZQ_3)/3,
185   M3h_LEPZQ_AVG = (M3h_LEPZQ_1 + M3h_LEPZQ_2 + M3h_LEPZQ_3)/3,
186   M12h_LEPZQ_AVG = (M12h_LEPZQ_1 + M12h_LEPZQ_2 + M12h_LEPZQ_3)/3,
187   M24h_LEPZQ_AVG = (M24h_LEPZQ_1 + M24h_LEPZQ_2 + M24h_LEPZQ_3)/3,
188   #now for the females
189   Fctl_LEPZQ_AVG = (Fctl_LEPZQ_1 + Fctl_LEPZQ_2 + Fctl_LEPZQ_3)/3,
190   F1h_LEPZQ_AVG = (F1h_LEPZQ_1 + F1h_LEPZQ_2 + F1h_LEPZQ_3)/3,
191   F3h_LEPZQ_AVG = (F3h_LEPZQ_1 + F3h_LEPZQ_2 + F3h_LEPZQ_3)/3,
192   F12h_LEPZQ_AVG = (F12h_LEPZQ_1 + F12h_LEPZQ_2 + F12h_LEPZQ_3)/3,
193   F24h_LEPZQ_AVG = (F24h_LEPZQ_1 + F24h_LEPZQ_2 + F24h_LEPZQ_3)/3,
194   #now mixed sex
195   Xctl_LEPZQ_AVG = (Xctl_LEPZQ_1 + Xctl_LEPZQ_2 + Xctl_LEPZQ_3)/3,
196   X1h_LEPZQ_AVG = (X1h_LEPZQ_1 + X1h_LEPZQ_2 + X1h_LEPZQ_3)/3,
197   X3h_LEPZQ_AVG = (X3h_LEPZQ_1 + X3h_LEPZQ_2 + X3h_LEPZQ_3)/3,
198   X12h_LEPZQ_AVG = (X12h_LEPZQ_1 + X12h_LEPZQ_2 + X12h_LEPZQ_3)/3,
199   X24h_LEPZQ_AVG = (X24h_LEPZQ_1 + X24h_LEPZQ_2 + X24h_LEPZQ_3)/3,
200   #average reps for the male LE at each timepoint
201   Mctl_LE_AVG = (Mctl_LE_1 + Mctl_LE_2 + Mctl_LE_3)/3,
202   M1h_LE_AVG = (M1h_LE_1 + M1h_LE_2 + M1h_LE_3)/3,
203   M3h_LE_AVG = (M3h_LE_1 + M3h_LE_2 + M3h_LE_3)/3,
204   M12h_LE_AVG = (M12h_LE_1 + M12h_LE_2 + M12h_LE_3)/3,
205   M24h_LE_AVG = (M24h_LE_1 + M24h_LE_2 + M24h_LE_3)/3,
206   #now for the females
207   Fctl_LE_AVG = (Fctl_LE_1 + Fctl_LE_2 + Fctl_LE_3)/3,
208   F1h_LE_AVG = (F1h_LE_1 + F1h_LE_2 + F1h_LE_3)/3,
209   F3h_LE_AVG = (F3h_LE_1 + F3h_LE_2 + F3h_LE_3)/3,
210   F12h_LE_AVG = (F12h_LE_1 + F12h_LE_2 + F12h_LE_3)/3,
211   F24h_LE_AVG = (F24h_LE_1 + F24h_LE_2 + F24h_LE_3)/3,
212   #now mixed sex
213   Xctl_LE_AVG = (Xctl_LE_1 + Xctl_LE_2 + Xctl_LE_3)/3,
214   X1h_LE_AVG = (X1h_LE_1 + X1h_LE_2 + X1h_LE_3)/3,
215   X3h_LE_AVG = (X3h_LE_1 + X3h_LE_2 + X3h_LE_3)/3,
216   X12h_LE_AVG = (X12h_LE_1 + X12h_LE_2 + X12h_LE_3)/3,
217   X24h_LE_AVG = (X24h_LE_1 + X24h_LE_2 + X24h_LE_3)/3,
218   #average reps for the male NMRI at each timepoint
219   Mctl_NMRI_AVG = (Mctl_NMRI_1 + Mctl_NMRI_2 + Mctl_NMRI_3)/3,
220   M3h_NMRI_AVG = (M3h_NMRI_1 + M3h_NMRI_2 + M3h_NMRI_3)/3,
221   M12h_NMRI_AVG = (M12h_NMRI_1 + M12h_NMRI_2 + M12h_NMRI_3)/3,
222   M24h_NMRI_AVG = (M24h_NMRI_1 + M24h_NMRI_2 + M24h_NMRI_3)/3,
223   #now for the females
224   Fctl_NMRI_AVG = (Fctl_NMRI_1 + Fctl_NMRI_2 + Fctl_NMRI_3)/3,
225   F3h_NMRI_AVG = (F3h_NMRI_1 + F3h_NMRI_2 + F3h_NMRI_3)/3,
226   F12h_NMRI_AVG = (F12h_NMRI_1 + F12h_NMRI_2 + F12h_NMRI_3)/3,
227   F24h_NMRI_AVG = (F24h_NMRI_1 + F24h_NMRI_2 + F24h_NMRI_3)/3,
228   #now mixed sex
229   Xctl_NMRI_AVG = (Xctl_NMRI_1 + Xctl_NMRI_2 + Xctl_NMRI_3)/3,
230   X30m_NMRI_AVG = (X30m_NMRI_1 + X30m_NMRI_2 + X30m_NMRI_3)/3,
231   X3h_NMRI_AVG = (X3h_NMRI_1 + X3h_NMRI_2 + X3h_NMRI_3)/3,
232   X12h_NMRI_AVG = (X12h_NMRI_1 + X12h_NMRI_2 + X12h_NMRI_3)/3,
233   X24h_NMRI_AVG = (X24h_NMRI_1 + X24h_NMRI_2 + X24h_NMRI_3)/3,
234   #for NMRI there were also Juvenile stages collected
235   Jctl_NMRI_AVG = (Jctl_NMRI_1 + Jctl_NMRI_2 + Jctl_NMRI_3)/3,
236   J30m_NMRI_AVG = (J30m_NMRI_1 + J30m_NMRI_2 + J30m_NMRI_3)/3,
237   J3h_NMRI_AVG = (J3h_NMRI_1 + J3h_NMRI_2 + J3h_NMRI_3)/3,
238   J12h_NMRI_AVG = (J12h_NMRI_1 + J12h_NMRI_2 + J12h_NMRI_3)/3,
239   J24h_NMRI_AVG = (J24h_NMRI_1 + J24h_NMRI_2 + J24h_NMRI_3)/3,
240   #now make columns of Log fold changes in gene expression between sexes for each strain
241   LogFC.FvsM_Ctrl_LEPZQ = (Fctl_LEPZQ_AVG - Mctl_LEPZQ_AVG),
242   LogFC.FvsM_Ctrl_LE = (Fctl_LE_AVG - Mctl_LE_AVG),
243   LogFC.FvsM_Ctrl_NMRI = (Fctl_NMRI_AVG - Mctl_NMRI_AVG),

```

dplyr::mutate
for very large dataframes

A grammar for making tables

